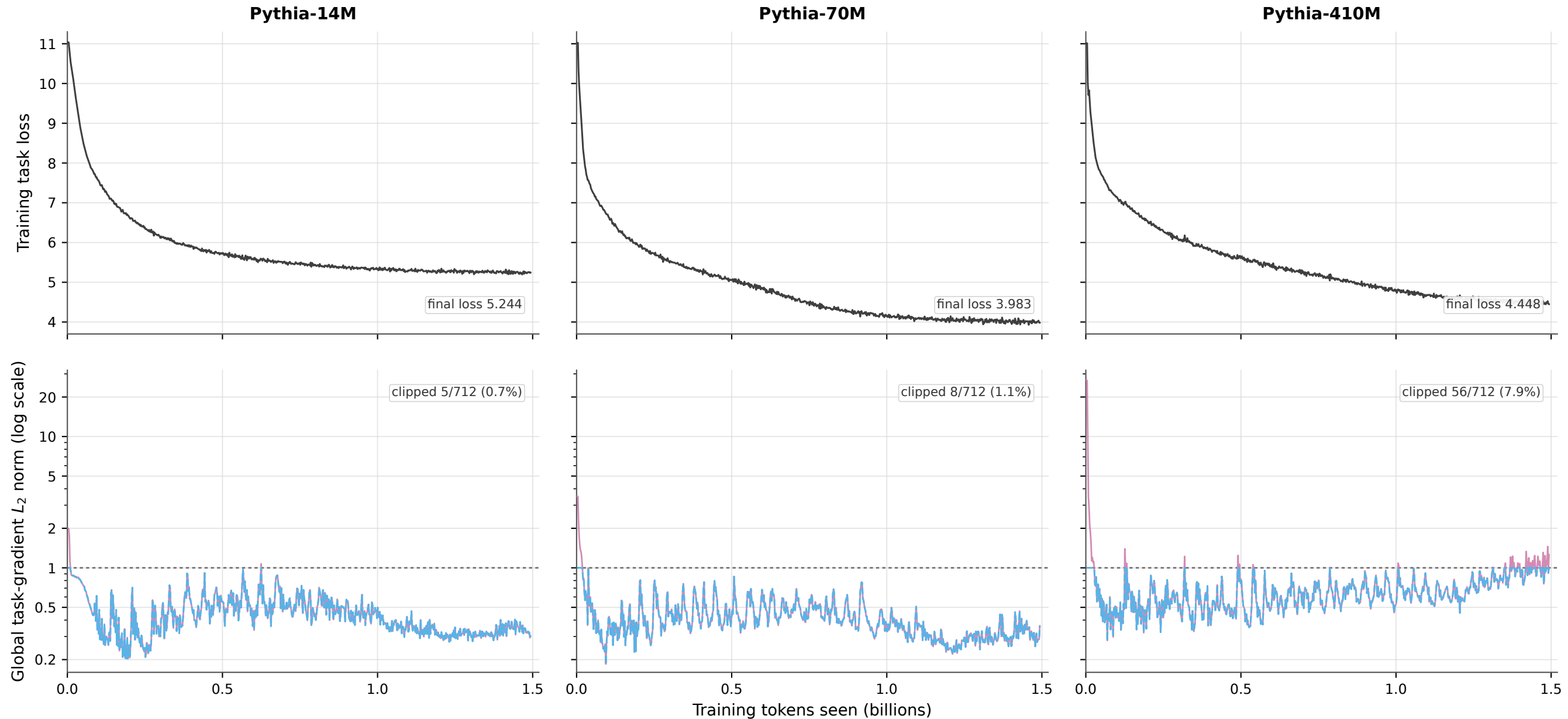


A0 optimization trajectories through one MiniPile pass

Training task loss and global full-model L2 gradient norm at every optimizer boundary

— Training task loss — Before clipping - - - After clipping - - - Global clip threshold = 1



Raw boundary values; no smoothing. Each boundary contains 2,097,152 tokens. All 712 updates completed without overflow or skipping.
The norm is not parameter-count normalized; identical global clipping at 1.0 therefore need not have identical effects across scales.